

Investigations of a Number of Alternative Negative Electrode Materials for Use in Lithium Cells

A. Netz, R.A. Huggins and W. Weppner

Faculty of Engineering, Christian-Albrechts University
Kaiserstr. 2, D-24143 Kiel, Germany

Abstract. There is a considerable interest in the replacement of graphite as the negative electrode reactant in rechargeable lithium batteries by composite electrodes containing alloys or convertible oxides. Some such materials can have much higher theoretical specific capacities than graphite, more than a factor of ten in some cases. In addition it would be desirable to eliminate the irreversible loss of capacity during the first charging cycle that is characteristic of graphite electrodes, as well to raise the operating potential somewhat in order to reduce the danger of the formation of elemental lithium during recharging.

The several strategies that have been followed in the search for attractive alternatives will be briefly described. It has been found to be difficult to obtain the desired combination of high capacity, low first cycle loss and capacity retention upon cycling.

Investigations of the electrochemical behaviour of elemental boron and borides (B_4C , CaB_6 , LaB_6 , AlB_2 , SiB_3), elemental silicon and silicides (Mg_2Si , $FeSi_2$, $CoSi_2$, $NiSi_2$, $TiSi_2$, VS_i_2) and of silicon monoxide, SiO , will be reported. The galvanostatic cycling method was used, with thick layer electrodes (30 μm) deposited upon copper foil in coffee bag-type cells with a liquid electrolyte. Lithium foil was used for the counter and reference electrodes. The results of the investigation of the morphological changes upon cycling, as observed by the use of SEM, will also be presented.

1. Introduction

1.1. General Considerations. The motivation of this study was the investigation of alternative negative electrode materials for rechargeable lithium ion batteries. Up to now carbon is used in commercial cells as anode material. In 1990 SONY announced the availability of lithium ion batteries based on hard carbon on the negative side and $LiCoO_2$ on the positive side [1]. The structure of carbon can be described as stacked closest-packed hexagonal layers. Lithium is able to intercalate between these planes up to a maximum concentration that corresponds to the formula LiC_6 [2,3]. Advantages for using carbon are a relatively flat operating potential of around 100 mV and reasonable cost. For safety considerations the potential may be too close to that of elemental lithium. The theoretical capacity of natural graphite is limited to 372

mAh/g, whereas in practical cells this value is not achieved. Other problems with carbon are the first cycle loss and the co-insertion of certain electrolytes, which leads to capacity loss upon the cycling.

Rechargeable batteries for applications in the electronics consumer market, like portable computers and cameras, handheld devices or mobile telephones, have a number of requirements. First they have to offer the desired power for a long period of time without recharging. Also these batteries should have reasonable lifetime upon cycling. The first point is related to the specific energy density and the latter to changes in the microstructure upon repeated charge-discharge cycles. It is obvious, that the energy density of carbon is limited, so there is a need in the replacement of carbon by other materials.

1.2. Alloys as Alternatives for Carbon. Dey showed in

1971 that lithium could alloy with metals in organic electrolytes by electrochemical reaction at ambient temperature [4]. Since then much effort has been done in this field. For the replacement of carbon by alloys some reasons should be pointed out. On the one hand side this materials should not show an irreversible loss, because due to the absence of oxygen. The theoretical capacities are much higher compared to LiC_6 , as shown in Table 1.

In case of silicon-alloy the capacity is more than a factor of ten from the value we can obtain for standard graphite. We will show later that high capacities are related to capacity fade upon cycling. This is because of large expansion in volume that occur upon reaction of tin with lithium. Cracks of the electroactive particles are the result of these volume changes and furthermore the fracture of particles, so that they can lose electronic contact to the current collector.

1.3. Materials Investigated in this Study. For this materials screening study the following groups of materials were investigated: elemental boron and borides, elemental silicon and silicides and silicon monoxide. In Table 2 all the investigated materials are listed.

The different structures of the materials we have chosen for this study are of interest. An excellent overview about this is given in the book of A. F. Wells [18].

In **elemental boron** all the B atoms belong to icosahedral groups B_{12} arranged in a cubic closest packing. The unit cell is rhombohedral, with B_{12} icosahedra placed at all the points of the rhombohedral lattice, as shown in Fig. 1.

Table 1. Gravimetric capacities for alloys of Li with aluminium [5,6,7], tin [8] and silicon [9,10,11].

Alloy Li-M	y Li / mol M	Capacity / mAh/g
Li-Al	0.9	815
Li-Sn	4.4	1000
Li-Si	4.4	4211

Table 2. Materials investigated in this study.

Boron / Borides	Silicon / Silicides	Siliconoxide
Boron	Silicon (crys.) [12]	SiO [17]
B_4C / CaB_6 / LaB_6	Silicon (amorph.)	
AlB_2	Mg_2Si [13-16]	
SiB_3	TiSi_2 / VSi_2	
	FeSi_2 / CoSi_2 / NiSi_2	

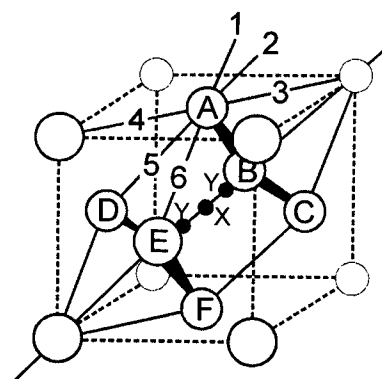


Fig. 1. Unit cell of elemental boron. B_{12} icosahedra are placed at all the points of the rhombohedral lattice.

Six B atoms of each icosahedron are linked to B atoms of another icosahedron. As a result we get a three dimensional network, in which each icosahedron is connected to six others and the corresponding bonds lie along the lattice directions. The other six B atoms of the icosahedron, which form a hexagon around its equator, are able to form three-centre-bonds in the plane ABC, which lies perpendicular to the body diagonal of the cell. This structure could serve as a basis for all the other borides. At the center of the cell is an octahedral hole, surrounded by the icosahedra, as shown in Fig. 1. Groups of atoms YXY can take place along the body-diagonal of the cell.

In case of boroncarbide B_4C this hole is filled up with linear CCC-chains and the three-centre-bonds in the equatorial plane are replaced by bonds between six B atoms of the B_{12} icosahedra and the terminal C atoms of the chains. So, one half of the B atoms form bonds to six B atoms and the other ones are linked to five B and one C. It can be seen that B_4C is isomorph to elemental boron and forms also a three-dimensional framework. The structure of CaB_6 and LaB_6 could be described as a simple cubic metal lattice, expanded by the introduction of octahedral B_6 groups. So, these two materials also consist of a framework structure.

In the case of AlB_2 the B atoms are arranged in graphite-like layers. Each B atom is equidistant from three other B atoms. The Al atoms are interleaved between the boron sheets in that way, that one boron is surrounded by six Al atoms at the vertices of a trigonal prism, as illustrated in Fig. 2.

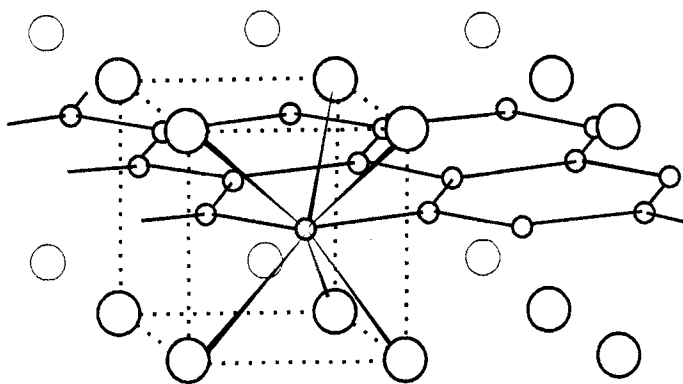


Fig. 2. Structure of AlB_2 . The smaller circles represent B atoms, the larger ones Al atoms.

The structure of SiB_3 is also isomorphous to elemental boron. Two Si atoms are sitting in the octahedral hole of the boron lattice. Some of the boron atoms are replaced by Si atoms in a statistical way, giving the composition SiB_3 . In this case the structure could also be described as a three-dimensional network.

Elemental silicon shows a diamond-like structure with tetrahedral arrangements of silicon bonds. Mg_2Si can be described as anti-fluorite structure with cubic closest packing, so that the silicon atoms are surrounded by 8 magnesium atoms, which occupy the vertices of a cube. TiSi_2 is built of close packed layers, with a ratio of 2:1 for Si to Ti. Distances of neighbour atoms are not equal, so that the Ti atoms are shifted from the layer. VSi_2 is isomorphous to TiSi_2 . FeSi_2 reveals a distorted fluorite structure, CoSi_2 and NiSi_2 a normal fluorite type one. It should be pointed out, that all the silicon related structures show a three dimensional network.

2. Experimental Setup

All electrochemical measurements were done in a coffee-bag type arrangement with thick layer working electrodes and elemental lithium as counter and reference electrodes. The measurements were done at ambient temperature (25°C).

2.1. Preparation of Electrodes. For preparing the working electrodes we used a mixture of 85 wt% electroactive material with 10 wt% polyvinylidene-difluoride (PVDF) as binder and 5 wt% acetylene black as conductive carbon. This mixture was ballmilled with a milling machine MM200 (RETSCH) to obtain fine and homogenous particle size. The ballmilled powder was slowly dispersed in 1-methy-2-pyrrolidone (NMP). This slurry was spread on rough copper foil in a thickness of $30\ \mu\text{m}$ with doctor

blade equipment. To evaporate the solvent in a moderate way we used an infrared light system (110°C for 15 hours) to avoid cracking of the film. After the film has dried the coated foil was cut into working electrodes of $2\times 3\ \text{cm}$ size. The amount of electroactive material on these electrodes were typically in the range of 10 to 30 mg, depending on the materials. Lithium foil (ALDRICH, 99.9 %) in a thickness of $380\ \mu\text{m}$ and $1\times 3\ \text{cm}$ size was used as counter and reference electrode. The lithium foil was cleaned in a glove box with argon atmosphere by scratching off surface layers.

2.2. Test Cell and Electrochemical Method. A schematic overview of our test cell is given in Fig. 3. All electrochemical measurements were done in this coffee-bag type cell, whereas the whole assembly of the cell was done in a glove box with argon atmosphere and oxygen/nitrogen purification. Copper foil of $1\times 5\ \text{cm}$ size was used as current collectors. The coffee-bag consists of a aluminium-polymer-laminate foil. As separator we used glassfiber (WHATMAN) in the size of $3\times 4\ \text{cm}$. The fiber was saturated with 1 mol/l LiPF_6 electrolyte in EC:DEC (2:1) solution from MERCK. After inserting the working electrode and the lithium foil, as shown in Fig. 3, the cell was sealed under vacuum to exhaust excess electrolyte and to get good contact of the electrodes to the current collectors. To prevent short circuiting due to the sealing, an isolating white tape was inserted at the top edge of the cell.

All cells were cycled in galvanostatic mode against lithium at a current density of $0.1\ \text{mA}/\text{cm}^2$. The voltage was monitored versus Li/Li^+ with a digital multimeter (KEITHLEY 195A). To apply and control the current a source measure unit (KEITHLEY 236) was used. The whole measurement system was controlled by a computer with LabView software.

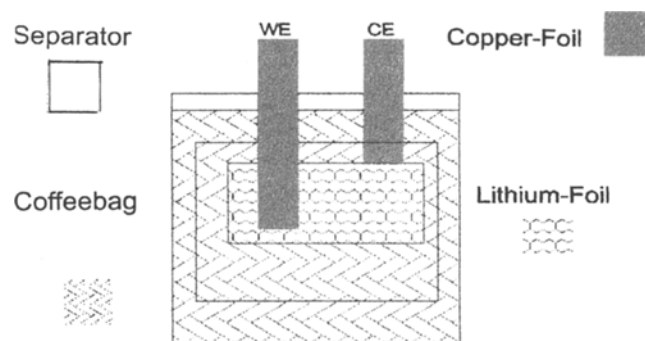


Fig. 3. Schematic overview of a coffee-bag type test cell. WE working electrode, CE counter electrode.

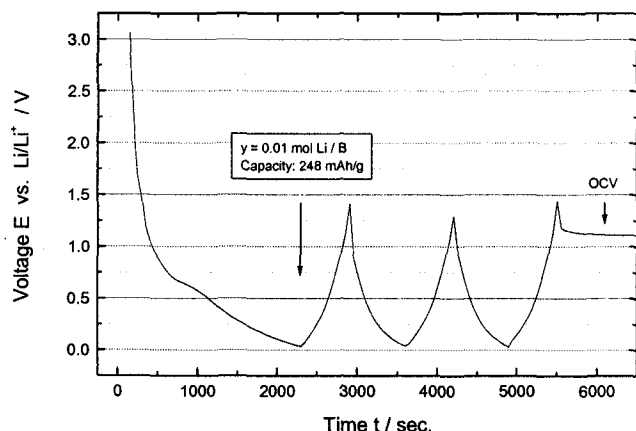


Fig. 4. Galvanostatic cycling of elemental boron. Three cycles are shown and the relaxation of the cell (open circuit voltage, OCV).

3. Experimental Results

3.1. Elemental Boron and Borides. In this chapter the results will be shown which we obtained for elemental boron and certain borides. Cycling was done between 25 mV and 1.5V, if not indicated otherwise.

Figure 4 shows a scan of galvanostatically cycled elemental boron. Up to 0.01 mol Li per mol of boron could be inserted, which corresponds to a first loading capacity of 248 mAh/g. It can be seen, that the cycleability is very poor with no significant amount of extractable lithium. The first loading capacities of B_4C , CaB_6 and LaB_6 are also very low in the range from 47 mAh/g for boron carbide, 21 mAh/g for calcium boride and 33 mAh/g for lanthanum boride. No extractable lithium was found with this materials and the voltage profiles are similar to that of elemental boron. In Fig. 5 galvanostatic cycling of AlB_2 is shown. This material exhibits a good cycleability with a poor reversible capacity of 22 mAh/g. Cycling was done between 25 mV and 2.5 V. The voltage profile for the insertion of lithium is noticeable different from the profile for the extraction, which indicates different reaction mechanisms. Again, the relaxation of the cell is shown after cycling. Beside the poor capacity the profile of the voltage is not favourable for an anode application.

For SiB_3 the capacities are much better. We have found a first loading capacity of 922 mAh/g and an extractable capacity of 440 mAh/g. In Fig. 6 the extractable lithium after initial loading of SiB_3 is shown. It can be seen that it is possible, to extract 1.0 mol from lithiated SiB_3 and insert the same amount of lithium in subsequent loading. The voltage profile is reasonably flat and the operating potential low, which is good for appli-

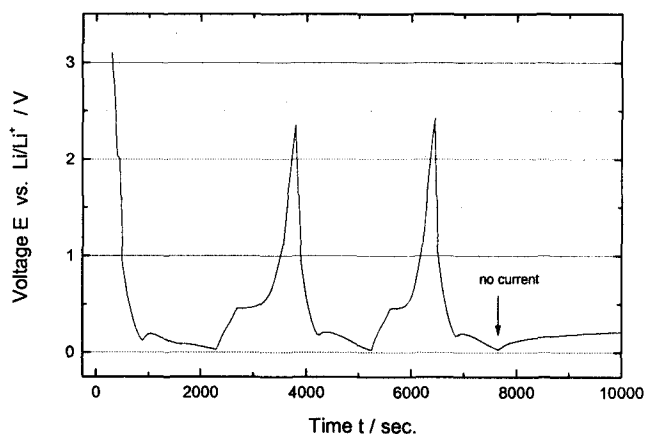


Fig. 5. Galvanostatic cycling of AlB_2 .

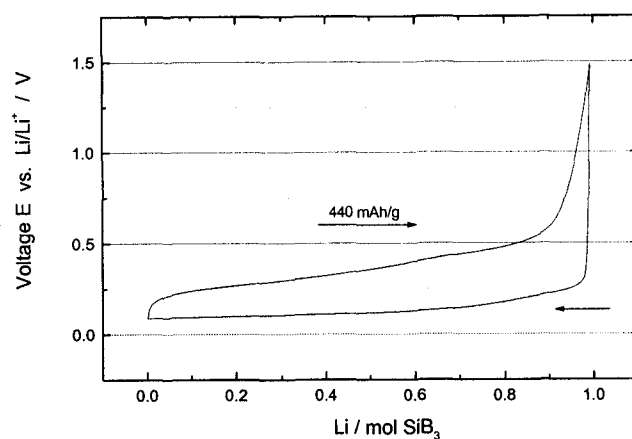


Fig. 6. Extractable lithium from SiB_3 after initial loading. Subsequent loading is also shown.

cation as anode material in a battery system. All data which we have obtained from galvanostatic measurements are collected in Table 3. Values are given for loading and extraction capacities of the first and third cycle.

3.2. Elemental Silicon and Silicides. We have performed measurements on elemental crystalline and amorphous silicon. In both cases we have found from galvanostatic cycling an unexpected first cycle loss. Up to 3.4 mol Li could be inserted into crystalline silicon and 0.4 mol Li is extractable, in that way 3.0 mol Li have been lost. In case of amorphous silicon 3.2 mol Li could be inserted and 1.1 mol Li could be extracted, which leads to a loss of 2.1 mol Li. In Fig. 7 the amount of extractable lithium from amorphous Si after initial loading is shown. Up to 1.1 mol Li is extractable from lithiated silicon. Similarities in the amount of extractable Li and voltage profile of elemental silicon to the previously shown SiB_3 are obvious. The voltage of the plateau region (operating poten-

Table 3. Results for elemental boron and certain borides from galvanostatic measurements.

Material	First loading / mAh/g	First extraction / mAh/g	Third loading / mAh/g	Third extraction / mAh/g
Boron	248	83	80	82
B ₄ C	47	16	17	16
CaB ₆	21	7	7	7
LaB ₆	33	11	10	10
AlB ₂	39	26	22	22
SiB ₃	922	440	322	246

tial) indicates the formation of Li-Si-alloy. In the further loading of the electrode, a loss can be seen. The extractable capacity decreases from 1023 mAh/g for the first cycle to 246 mAh/g for the third cycle.

REM pictures of the cycled silicon electrode have been taken in order to find the reason for this capacity loss upon cycling. As can be seen in Fig. 8 the particles are cracked and have lost the contact to the copper current collector. The coating is no more mechanically stable on the collector.

The electrochemical behaviour of Mg₂Si is shown in Fig. 9. We have found, that 1.3 mol of Li could be extracted from the sample after an initial loading of 1.9 mol Li, which is a first cycle loss of 0.6 mol Li. In subsequent loading 1.5 mol Li could be inserted again. The voltage profile is not flat. During extraction of lithium four plateaus at 182, 280, 440 and 650 mV could be identified, as can be seen in Fig. 9. These plateaus in the voltage profile are related to phase transformations of the sample upon delithiation.

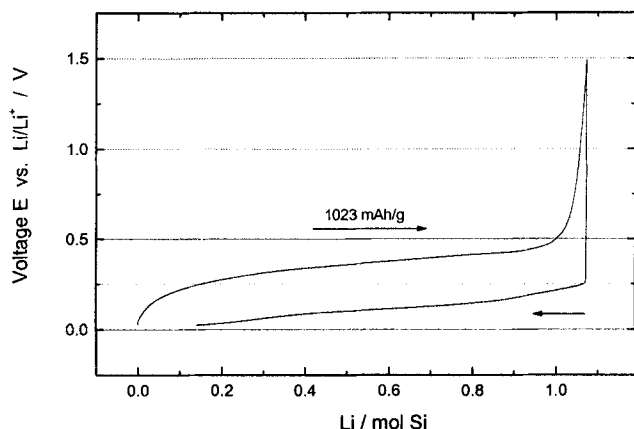


Fig. 7. Extractable lithium from lithiated amorph. silicon and subsequent loading.

In the case of FeSi₂ we have obtained a first loading of 0.34 mol Li and an extractable amount of 0.25 mol Li

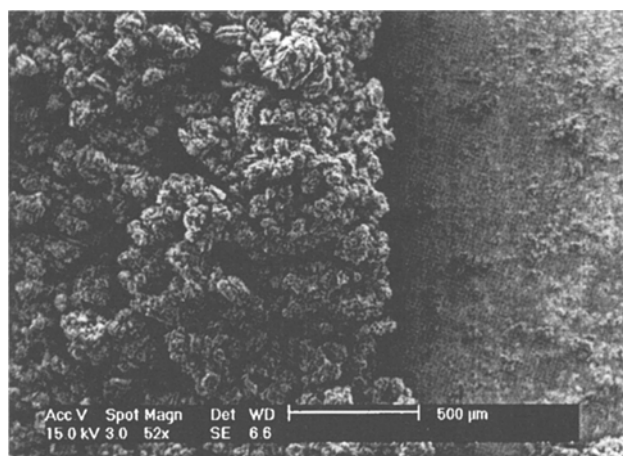
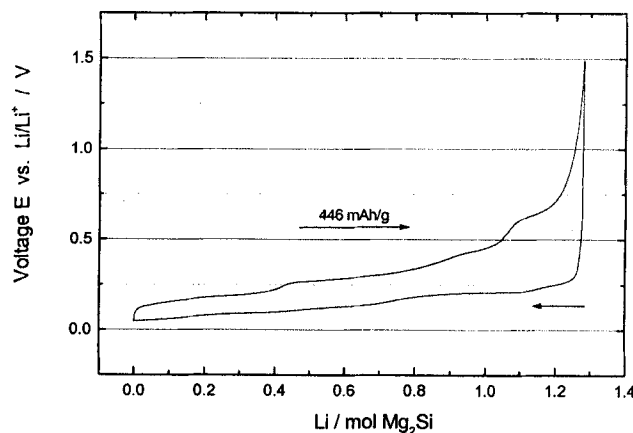


Fig. 8. REM image of a cycled silicon working electrode. The particles are cracked and have lost contact to the current collector.

Fig. 9. Extractable lithium from lithiated Mg₂Si and subsequent loading.

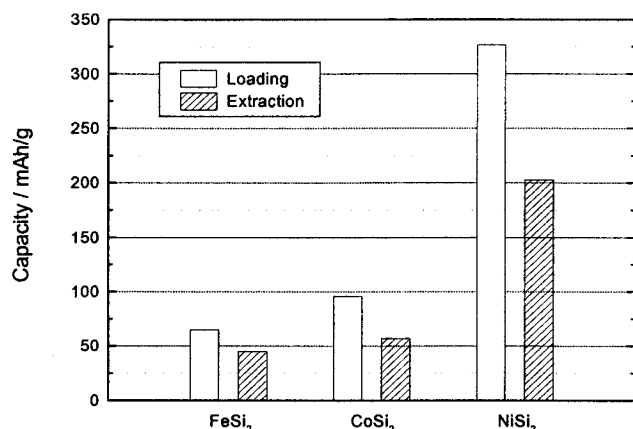


Fig. 10. Loading and extraction capacities for FeSi₂, CoSi₂ and NiSi₂.

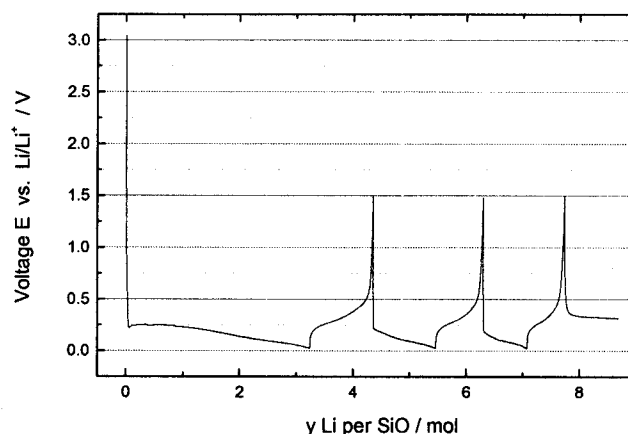


Fig. 11. Galvanostatic cycling of SiO.

during subsequent delithiation. For CoSi₂ we can insert 0.41 mol Li and extract 0.25 mol Li. For NiSi₂ we have obtained a first loading of 1.4 mol Li and an extraction of 0.87 mol Li. Figure 10 shows the first loading capacities and extraction capacities for the whole group. It can be seen, that for these materials the amount of insertable and extractable lithium increases due to their positions in the periodic system in the elements. All results we obtained for silicon and certain silicides are given in Table 4.

3.3. Silicon Monoxide. Galvanostatic cycling of SiO is shown in Fig. 11. Three cycles are shown. The typical first cycle loss of 2 mol of Li is obvious, which is related to the reaction of lithium with the oxide during first loading to form Li₂O. Around 1 mol of lithium is extractable again, as can be seen in Fig. 12.

The profile of the voltage after the first loading is similar to that we obtained for elemental silicon, so we assume the formation of elemental silicon during the first cycling step.

4. Summary and Conclusions

4.1. Elemental Boron and Borides. At ambient temperature the amount of lithium, which could react with elemental boron is in the order of 0.01 mol. We have not obtained any significant cycleability of this material. The same results we have found for the borides B₄C, CaB₆ and LaB₆. The only layered compound in this group, AlB₂, shows a good cycleability, but a reversible capacity of only 22 mAh/g. Only the SiB₃ has offered a reversible capacity of 440 mAh/g with a flat operating potential. We are still working in reducing the first cycle loss and improving the cycleability of siliconboride, which is a matter of paste optimization. Similarities of the voltage profile of the cycled SiB₃ with the results we obtained for elemental silicon are obvious. We assume the formation of silicon during the first loading step. The further reversible reaction is determined to the formation of lithium silicon alloy.

4.2. Elemental Silicon and Silicides. The big first cycle loss which we have found for crystalline and amorphous

Table 4. Results for elemental silicon and certain silicides from galvanostatic measurements.

Material	First Loading / mAh/g	First Extraction / mAh/g	Third loading / mAh/g	Third extraction / mAh/g
Si (crys.)	3230	413	157	132
Si (amorph.)	3003	1023	356	246
Mg ₂ Si	703	583	575	450
TiSi ₂	27	11	10	11
VS ₂	0	0	0	0
FeSi ₂	65	45	50	47
CoNi ₂	96	57	44	40
NiSi ₂	327	203	129	112

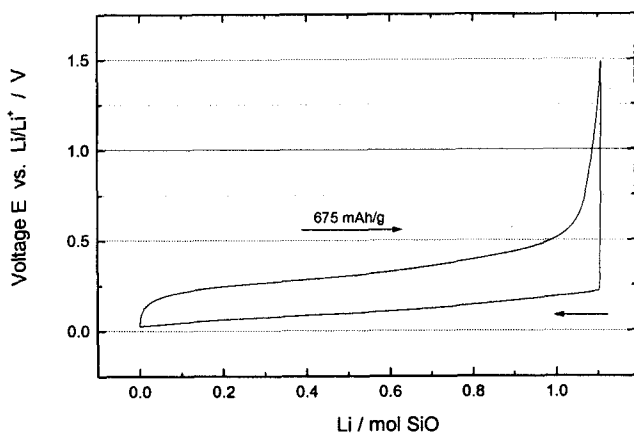


Fig. 12. Extractable lithium from lithiated SiO and subsequent loading.

silicon is unexpected. REM pictures of the electrodes after cycling show cracks and fragmented particles with no more contact to the current collector. So also for this material the optimization of the paste for coating will be done, in order to see, if this loss is due to coating problems. The alternative would be trapped lithium into the structure of silicon. Mg_2Si is of interest because of the reversible capacity of 583 mAh/g. But the voltage profile reveals four plateaus, which indicate phase transformations of the sample. TiSi_2 and VSi_2 do not react with lithium. In the group of FeSi_2 / CoSi_2 / NiSi_2 we have found an increase in loading and extraction capacity, due to their positions in the periodic system of the elements. For application in battery systems the capacities are too small.

4.3. Silicon Monoxide. For the silicon monoxide the obtained extraction capacity of 635 mAh/g is fine, but the first cycle loss of 2 mol Li is still a problem. Also in this silicon-based compound we assume the formation of elemental silicon during the first lithiation and further reaction of this in-situ generated silicon with lithium. Voltage profiles of the sample, which are similar to those we got from both elemental silicons, give evidence to this assumption.

5. Acknowledgement.

This work was supported by the European Commission under the Joule Program, Contract No. JOR3-CT98-0180.

6. References

- [1] T. Nagaura, K. Tozawa, *Prog. Batteries Sol. Cells* **9**, 209 (1990).
- [2] N. Imanishi, S. Ohashi, T. Ichikawa, T. Takeda and O. Yamamoto, *J. Power Sources* **39**, 85 (1992).
- [3] R. Kanno, Y. Takeda, T. Ichikawa, K. Nakanishi and O. Yamamoto, *J. Power Sources* **26**, 535 (1989).
- [4] A.N. Dey, *J. Electrochem. Soc.* **118**, 1547 (1971).
- [5] C.J. Wen, B.A. Boukamp, W. Weppner and R.A. Huggins, *J. Electrochem. Soc.* **126**, 2258 (1979).
- [6] N.P. Yao, L.A. Heredy and R.C. Saunders, *J. Electrochem. Soc.* **118**, 1039 (1971).
- [7] E.C. Gay, D.R. Visers, F.J. Martino and K.E. Anderson, *J. Electrochem. Soc.* **123**, 1591 (1976).
- [8] J. Wang, I.D. Raistrick and R.A. Huggins, *J. Electrochem. Soc.* **133**, 457 (1986).
- [9] A. Anani, S. Crouch-Baker and R.A. Huggins, *J. Electrochem. Soc.* **134**, 3098 (1987).
- [10] R.N. Seefurth and R.A. Sharma, *J. Electrochem. Soc.* **124**, 1207 (1977).
- [11] R.N. Seefurth and R.A. Sharma, *J. Electrochem. Soc.* **127**, 1101 (1980).
- [12] W. Weydanz, PhD Thesis, Univ. of Ulm (1997).
- [13] R.A. Huggins and A.A. Anani, US Patent 4.950.566, Aug.21 (1990).
- [14] A.A. Anani and R.A. Huggins, *J. Power Sources* **38**, 363 (1992).
- [15] C.-K. Huang, S. Surampudi, A.I. Attia and G. Halpert, US Patent 5.294.503, Mar. 15 (1994).
- [16] J. Santos-Pena, T. Brousse and D.M. Schleich, *Ionics* **6**, 1 (2000).
- [17] B. Klausnitzer, PhD Thesis, Univ. of Ulm (2000).
- [18] A.F. Wells, *Structural Inorganic Chemistry*, 5th edition (1984), Oxford University Press.

Paper presented at the 7th Euroconference on Solid State Ionic, Calcatoggio, Corsica, France, Oct. 1-7, 2000.

Manuscript rec. Apr. 20, 2001; acc. Juli 15, 2001.